

# A Comparative Evaluation of Structural MRI Foundation Models for Age, Sex, and Body-mass Index Predictions

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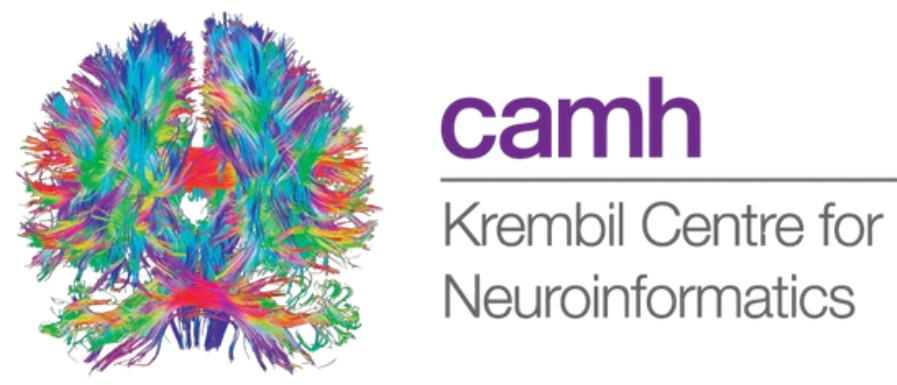
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Add a model to  
BrainFMBench!

Open a PR: [github.com/epics-lab/BrainFMBench](https://github.com/epics-lab/BrainFMBench)



## Motivation

Foundation models pretrained on large neuroimaging datasets could offset the small samples typical of clinical studies. Whether their representations generalize across populations and beat standard morphometric features remains untested.

## Methods

- Models: AnatCL (global and local descriptors), BrainIAC, 3D-Neuro-SimCLR, SwinBrain
- Baseline: FreeSurfer cortical thickness and surface area per brain region, from the Desikan-Killiany (aparc) and Schaefer 400 atlases
- Datasets: Parkinson's Progression Markers Initiative (PPMI,  $n = 894$ , older adults), Nathan Kline Institute (NKI,  $n = 958$ , lifespan), Healthy Brain Network (HBN,  $n = 1,000$ , youth)
- Tasks: sex classification, brain age, and body mass index (BMI); errors are mean absolute error (MAE)
- Probe: frozen features, identical Random Forest for every model, 90/10 train/test split, 5 fold cross validation, 5 random seeds
- Preprocessing: TurboPrep for all models except AnatCL, which uses CAT12; BrainIAC and 3D-Neuro-SimCLR were not tested on PPMI because they were pretrained on it

## Robustness

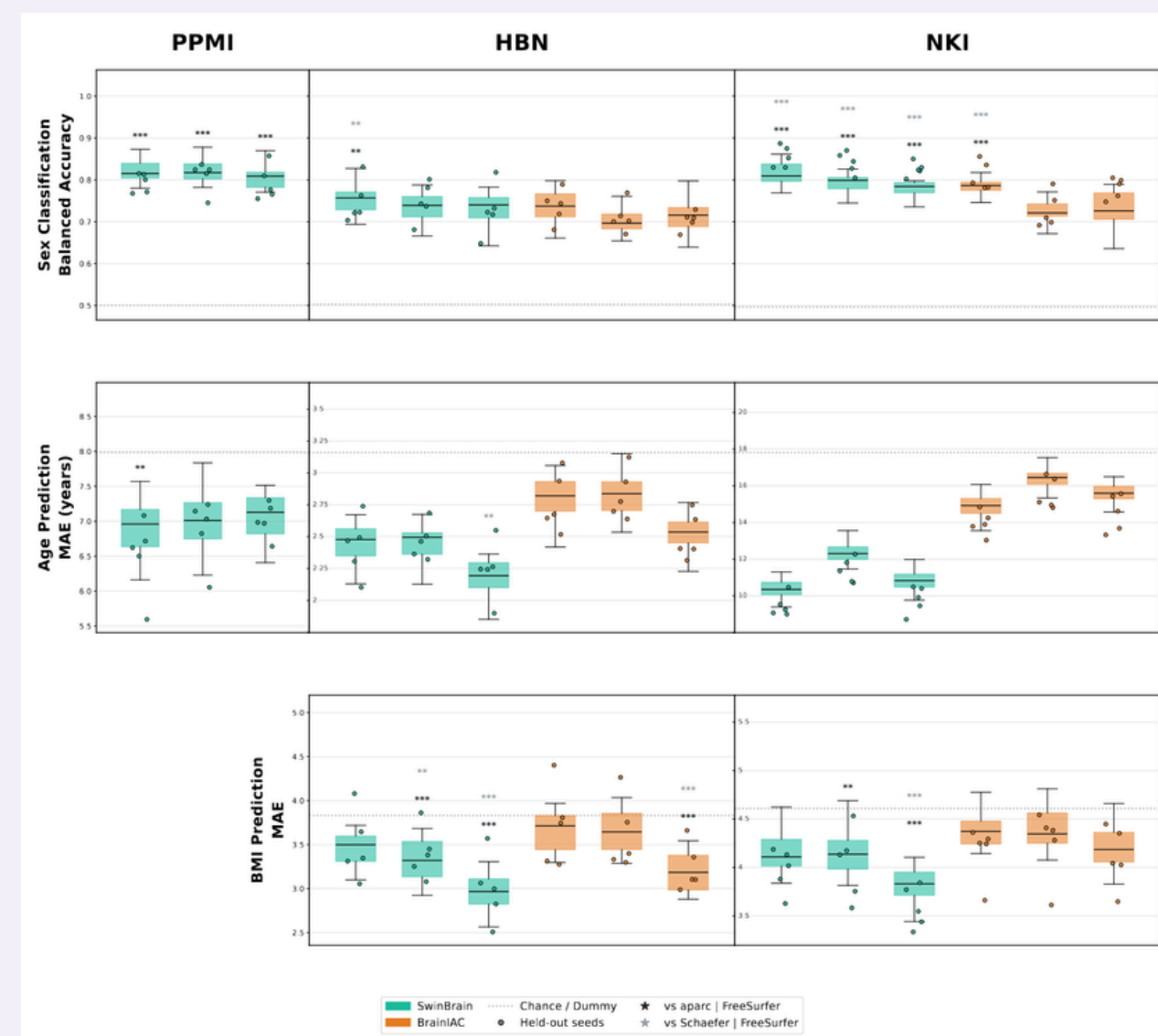


Figure 3: BrainIAC and SwinBrain under TurboPrep, T1 template registration, and the BrainIAC reference template.

BrainIAC and SwinBrain results were stable across three preprocessing pipelines, suggesting TurboPrep did not favor 3D-Neuro-SimCLR (which was not itself retested).

## Performance

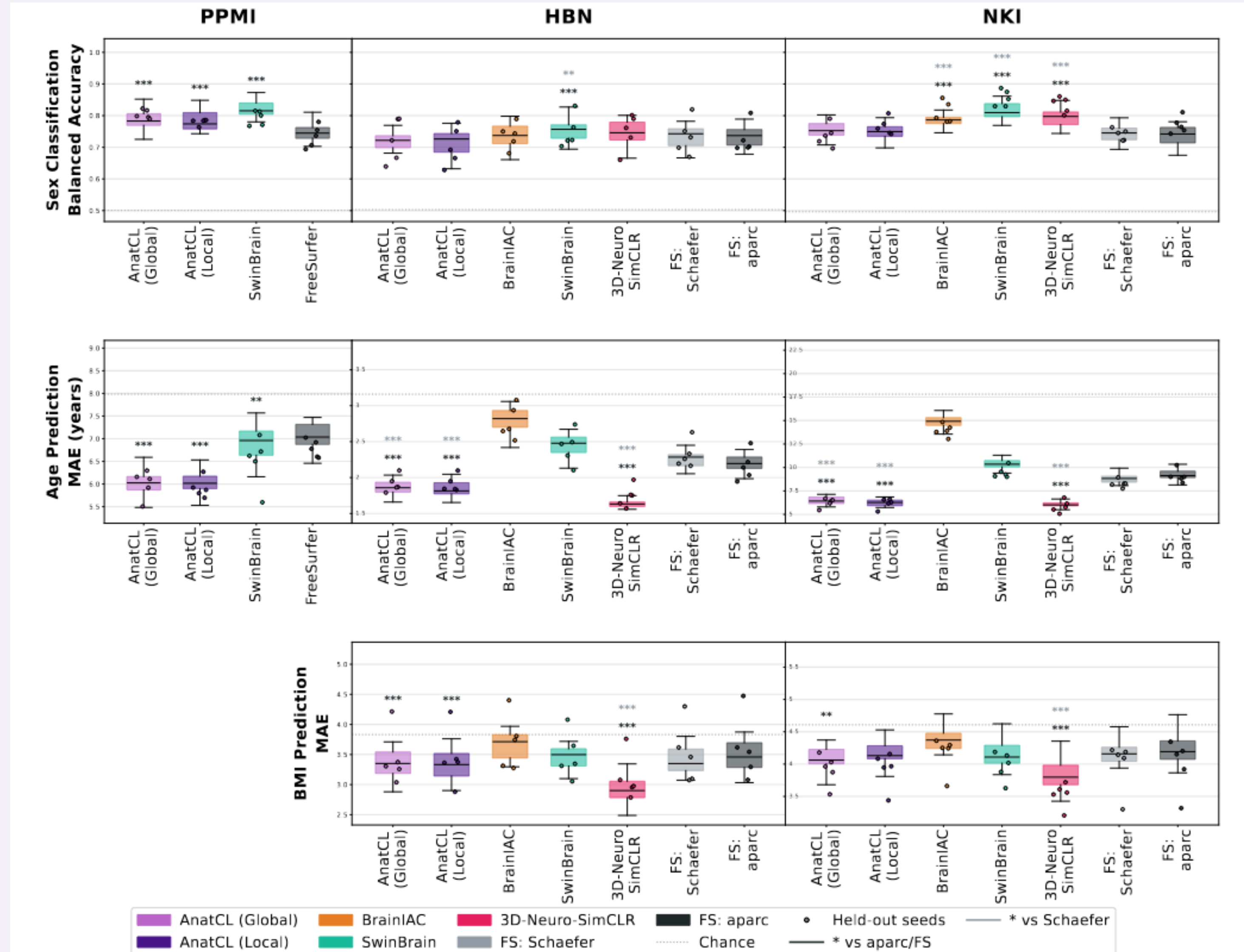


Figure 1: Each box shows 5 fold cross validation scores across 5 random seeds (25 values); dots are held out test results. Higher balanced accuracy and lower MAE are better. FS: aparc and FS: Schaefer are the FreeSurfer baselines (cortical thickness and surface area from the Desikan-Killiany and Schaefer 400 atlases; PPMI uses aparc only). Asterisks: Bonferroni corrected permutation tests vs FreeSurfer (\*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ ).

- 3D-Neuro-SimCLR and AnatCL outperformed FreeSurfer overall.
- 3D-Neuro-SimCLR beat both baselines on every HBN and NKI task except HBN sex classification.
- AnatCL gave the lowest brain age MAE on PPMI and beat FreeSurfer on age in all cohorts.
- BMI was hardest: only 3D-Neuro-SimCLR beat both baselines, MAE 3.11 vs 3.51 in HBN and 3.52 vs 3.99 in NKI.
- BrainIAC and SwinBrain did not consistently beat baselines.

## Takeaways

- The two best models share a ResNet-18 backbone; the two transformers did not reliably beat baselines.
- Pretraining data may matter: 3D-Neuro-SimCLR was trained only on T1 weighted scans, matching our evaluation data.
- Scope: frozen probe only, no general vision encoder baselines, within dataset evaluation.

## Representation alignment

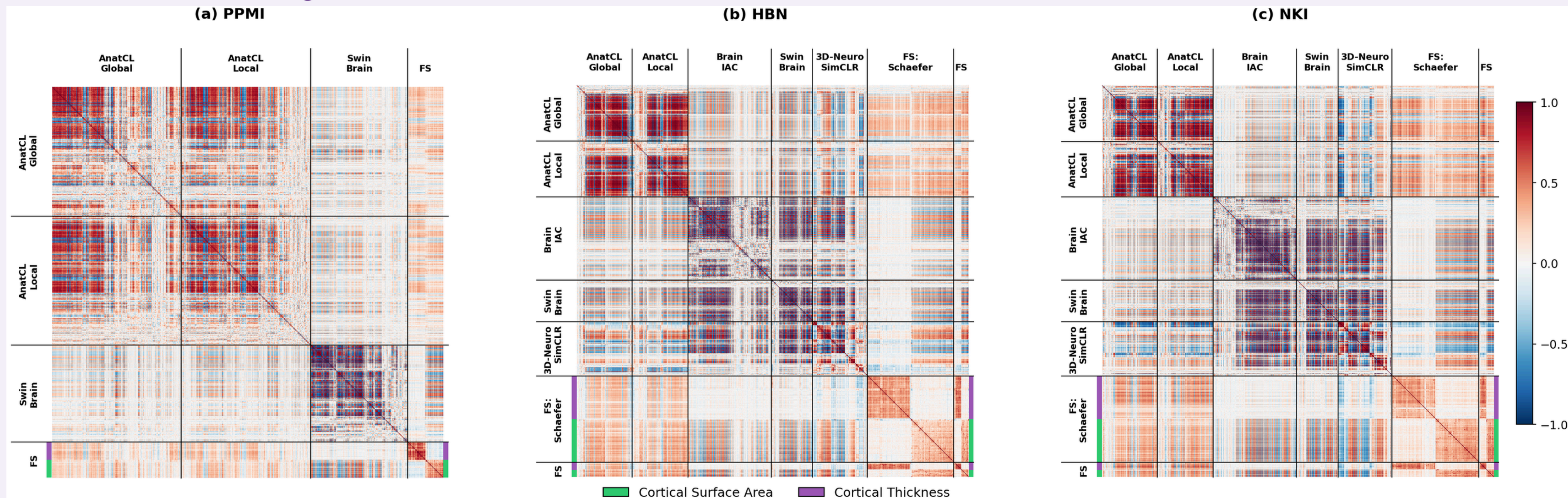


Figure 2: Pearson correlations between every pair of feature dimensions across all models in (a) PPMI, (b) HBN, and (c) NKI. Features within each model are reordered by hierarchical clustering (Ward's method); FreeSurfer features are colour coded (green: cortical surface area, purple: cortical thickness).

**Interpretation:** rows and columns are grouped by model, so each off diagonal block shows cross model correlations. Warm colours indicate positive and cool colours negative correlation across participants. Coloured bars mark FreeSurfer surface area (green) and thickness (purple) features.

AnatCL global and local are its two descriptors: local captures region specific variation, global aggregates features across the whole brain.

A higher  $|r|$  means a model's features carry information similar to that FreeSurfer measure. BrainIAC, SwinBrain, and 3D-Neuro-SimCLR features track cortical surface area (mean  $|r| \approx 0.33$  to  $0.37$ ), not thickness (mean  $|r| \approx 0.06$  to  $0.12$ ). AnatCL correlates equally with both (mean  $|r| \approx 0.25$ ), consistent with its pretraining on anatomical measures.



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### A note from the first author

I was unable to attend in person due to visa circumstances. Questions and collaborations are very welcome: arel.encin@camh.ca

Data: PPMI and the Reproducible Brain Charts  
initiative. Compute: Calcul Québec and the Digital  
Research Alliance of Canada.

### Key references

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